Mobile Banking Trends & Performance Analysis (2016 to 2025)

1. Project Objective:

The objective of this mini project is to perform real-world financial data analysis using:

- **Microsoft Excel** for data cleaning, transformation, and preparation
- **Power BI** for interactive dashboard creation and insight generation

This project focuses on analyzing **mobile banking statistics** to assess:

- Transaction volume and value distribution
- Customer activity and engagement levels
- Bank-wise performance (Public, Private, Co-operative)
- Transaction frequency and average value trends
- Risk indicators and adoption patterns

The goal is to transform raw mobile banking transaction data into **meaningful insights for decision-making**, highlighting customer behavior, banking sector performance, and digital adoption trends.

2. Data Sources:

- **Sources**: India Data Portal https://indiadataportal.com/p/reserve-bank-of-india/r/rbi-bankwise_mobile_banking_transactions-in-mn-aaa
- **Timeline**: January 2016 to December 2025
- **Domain**: Financial Analytics

3. Problem Statement:

The rapid growth of mobile banking makes it important to analyze transaction data and customer usage patterns.

This project focuses on evaluating **Mobile banking performance** using metrics like transactions, transaction value, and active customers across different states and bank types.

It also aims to identify **trends and customer behavior** to improve banking services and support better decision-making.

Bank Performance Analysis

To measure overall bank performance using transactions, transaction value, and active customers.

- To find top and low performing banks based on transaction volume and value.
- To compare performance across Public, Private, and Cooperative banks.
- To understand customer distribution across different banks.
- To check efficiency using average transaction value and transaction frequency.
- To understand each bank's share in total mobile banking activity.

Customer Behavior & Regional Insights

- To study customer distribution across states and identify adoption levels.
- To understand customer segments (Small, Medium, Large) and their contribution to transactions.
- To observe transaction patterns across different states and categories.
- To identify states with highest and lowest mobile banking usage.
- To compare spending behavior using average transaction value across states.

Trend & Time Analysis

- To study transaction trends over time and overall growth pattern.
  - To measure Year-over-Year (YoY) growth in transactions.
  - To observe monthly trends and seasonal changes in usage.
  - To identify long-term patterns for future forecasting.
  - To compare performance over different time periods using historical data.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
id	Integer	Unique identifier for each record
Date	Date	Transaction date
Month	String	Month of the transaction
Year	Integer	Year of the transaction
bank_name	String	Name of the bank
State	Categorical	State where the bank operates
no_of_transactions	Integer	Total number of transactions
amt_of_transactions	Decimal	Total transaction amount
no_of_active_customers	Integer	Number of active customers
Transaction Category	Categorical	Type of transaction (Small / Medium / Large)
Bank Type	Categorical	Type of bank (Public / Private / Co-operative)
Average transaction value	Decimal	Average value per transaction
Customer Tier	Categorical	Customer segment (Small / Medium / Large)
Transaction Frequency per Customer	Integer	Average transactions per customer

5. Tools & Technologies:

Microsoft Excel & Power Query:

- Excel
- Power Query
- Power BI

6. Data Pre-Processing (Excel / Power Query/ Power BI):

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicate records, handled missing values, standardized formats, and ensured correct data types for all columns.
- **Handling Missing Values:** Imputed missing values in no_of_active_customers using bank-wise logic to maintain data consistency and accuracy.
- **Power Query Transformations:** Applied transformations such as changing data types, creating custom columns, and automating data preparation steps.
- **Feature Engineering (New Columns Added):**
  Created additional columns such as:
- **Transaction Category:** Classified transactions into small, medium, and large categories.
- **State:** Added a column to capture the state where the bank operates.
- **Month and Year:** Extracted from the Date column to enable time-based analysis.
- **Average Transaction Value:** Calculated the average value per transaction.
- **Customer Tier:** Segmented customers into small, medium, and large tiers.
- **Transaction Frequency per Customer:** Derived the average number of transactions per customer.
- **Calendar Table Creation:** Created a separate date table to enable time-based analysis such as Year, Month, and support time intelligence functions.
- **Time Intelligence:** Implemented measures like YoY Growth % and MoM Growth % to analyze performance trends over time.
- **Filtering & Sorting:** Organized and filtered data to focus on relevant records for analysis.

- **Bank-wise Aggregation:** Created a calculated table to derive bank-wise transaction counts, enabling efficient comparison of bank performance.

7. Fact and Dimension Table Creation:

Structured the dataset into:

- **Fact Table:** Mobile Banking Statistics
- **Dimension Tables:** Calendar Table

7.1 Classification of Fact and Dimension Tables:

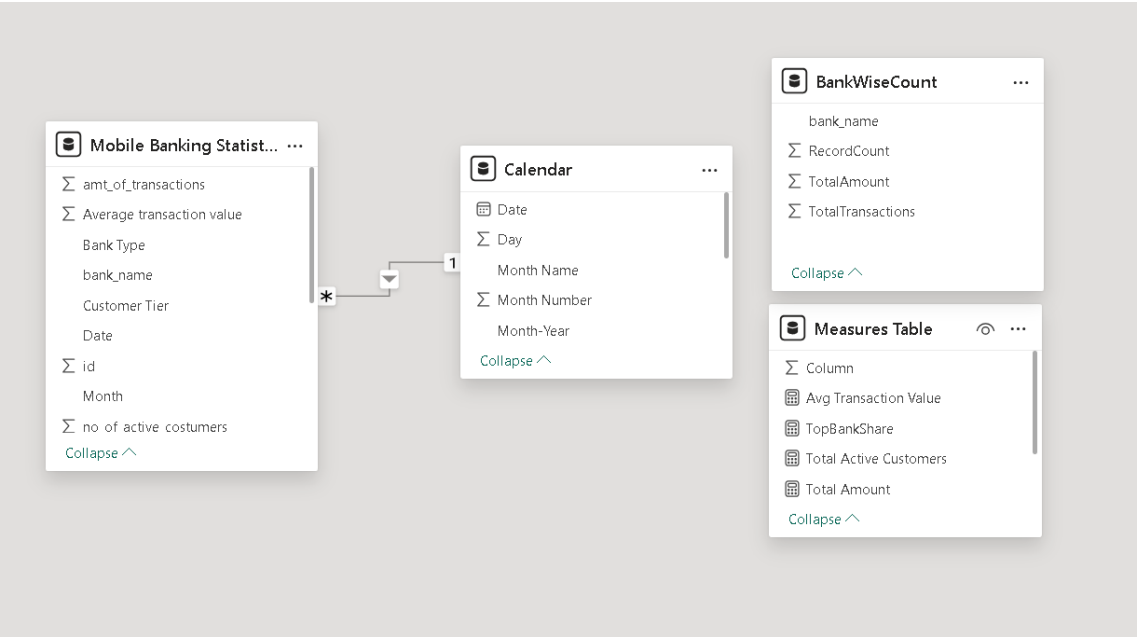
The mobile banking dataset was structured using a **star schema model**.

- The **Mobile Banking Statistics table** was treated as the **Fact Table** since it contains measurable attributes such as the number of transactions, average transaction value, number of active customers, and customer tier.
- The **Calendar table** was treated as a **Dimension Table** because it provides descriptive time attributes such as year, month, and quarter, which are essential for filtering and trend analysis.

8. Data Modelling and DAX (Power BI):

Data Model

- Established relationships between the **Mobile Banking Statistics table** and the **Calendar table**.
- Defined **one-to-many relationships** between the Calendar table and the fact table.
- Ensured proper **cardinality** for accurate aggregation and filtering.



9. DAX Measures & Calculated Columns:

Calculated Measure:

Implemented DAX measures for key metrics, including:

- **Total Transactions** – Represents the total number of transactions
- **Total Amount** – Represents the overall transaction value
- **Total Active Customers** – Shows the number of customers actively performing transactions
- **Total Customers** – Represents the total number of customers in the dataset
- **Total Banks** – Represents the total number of unique banks
- **Transaction Value** – Indicates the average value per transaction
- **Growth Percentage** – Measures the percentage change between current and previous period
- **YoY Growth %** – Measures the year-over-year percentage change in performance
- **Cumulative Transactions** – Displays the running total of transactions over time

Calculated Columns:

- **Month:** Represents the month of the transaction.
- **Year:** Represents the year of the transaction.
- **State:** Identifies the state where the bank operates.
- **Transaction Frequency per Customer (Integer):** Calculates the average number of transactions per customer.
- **Customer Tier:** Segmented customers into small, medium, and large tiers.
- **Transaction Frequency per Customer:** Derived the average number of transactions per customer.
- **Data Categorization:**
  Classified data based on Transaction Category, Bank Type, and Customer Tier for better segmentation and analysis.

Table Creation:

1. Built a **Calendar table** with attributes like Year, Month, Quarter, and Day to enable time-based filtering and trend analysis.

Relationship Establishment

- A **Many-to-One relationship** was established between:
- **Mobile Banking Statistics [Date] → Calendar [Date]**
- This relationship enabled:
- Monthly trend analysis
- Year-wise comparison
- Drill-down functionality
- Time intelligence calculations

2. Created a **BankWiseCount table** to store aggregated bank-level information such as Record Count, Total Transactions, and Total Amount.

10. Analysis and Visualizations (Power BI):

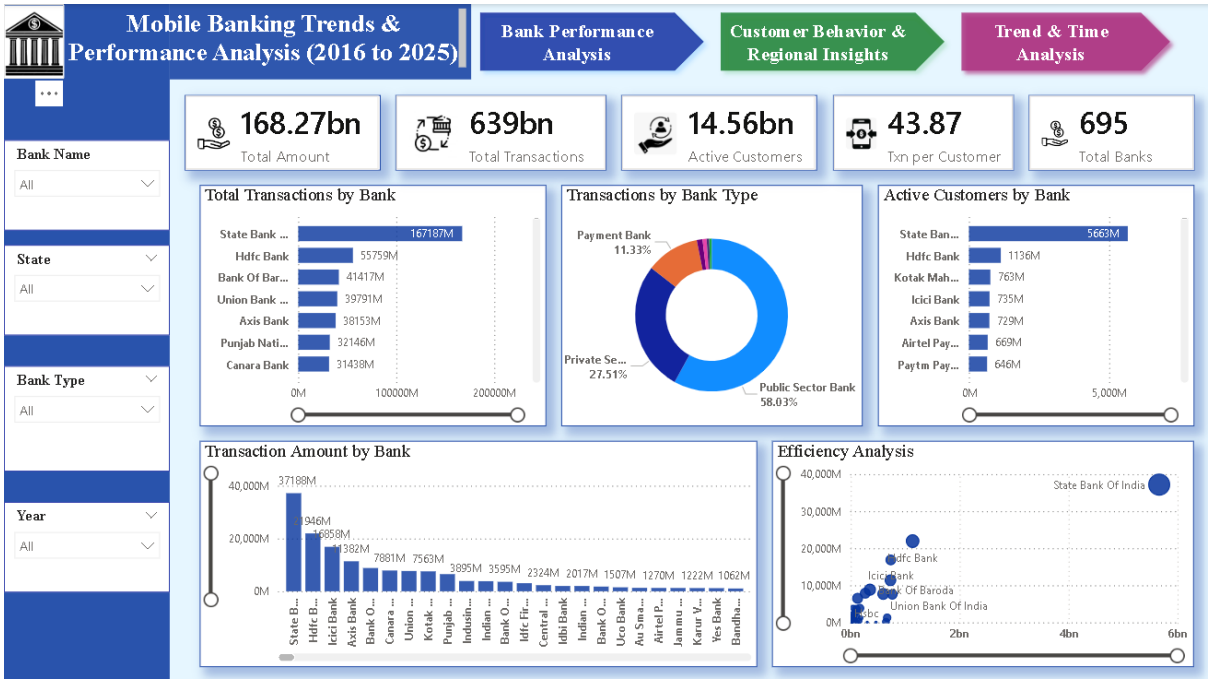
Bank Performance Analysis

Visuals Used:

- **KPI Cards** (Total Transactions, Total Transaction Amount, Active Customers, Transaction per Customer, Total Banks).
- **Bar Chart** (Transactions by Bank)
- **Column Chart** (Transaction Amount by Bank)
- **Donut Chart** (Bank Type Distribution)
- **Scatter Plot** (Active Customers vs Transaction Amount – Efficiency Analysis).
- **Page Navigator**

Interactive Features:

- **Slicers** (Bank Type, Bank Name, State, Year)
- **Button Navigation** (custom buttons for page switching)
- **Edit Interactions** (controlled slicer impact on visuals and cards)
- **Consistent Layout and Color Theme**



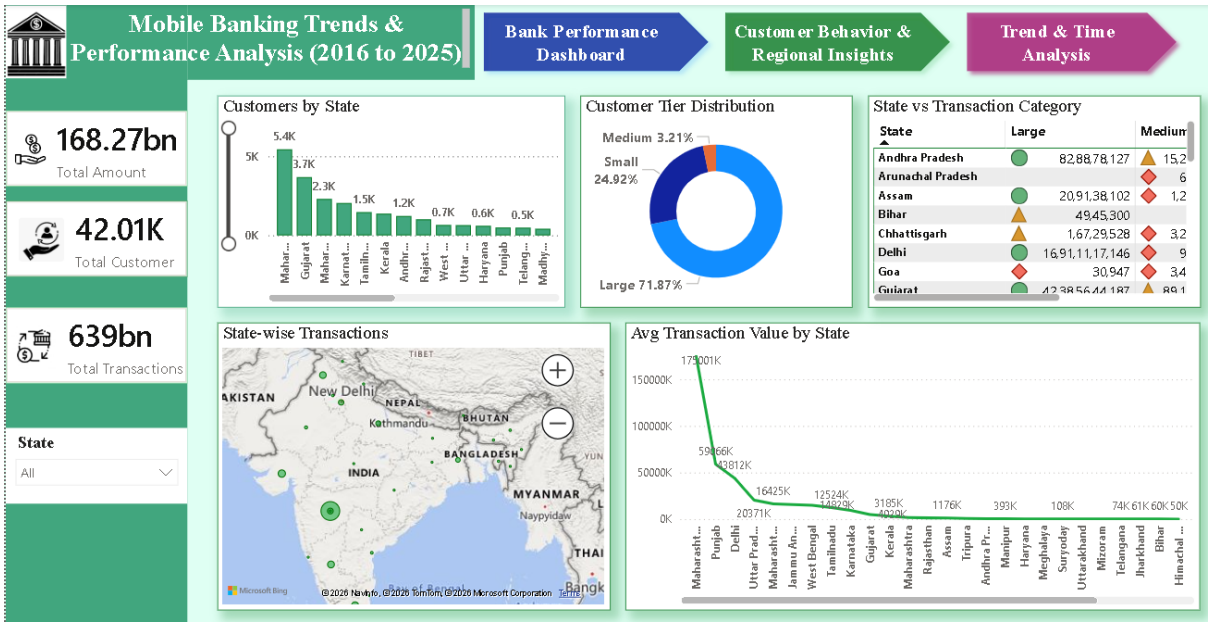
Customer Behavior & Regional Insights

Visuals Used:

- **Bar Chart (Customers by State) –** shows regional customer distribution
- **Donut Chart (Customer Tier Distribution) –** displays segmentation (Small / Medium / Large).
- **Matrix / Heatmap (State vs Transaction Category) –** highlights usage patterns across regions.
- **Map (State-wise Transactions) –** visualizes geographic transaction distribution
- **Line / Column Chart (Avg Transaction Value by State) –** shows spending behavior across states.

Interactive Features:

- **Slicers (State, Transaction Category) –** for regional and behavioural filtering
- **Slicers (Year) –** To see the year wise bank performance
- **Minimalistic Design –** avoids clutter and focuses on insights
- **Cross-filtering between visuals –** selecting one chart updates others
- Consistent Color theme and layout



Trend & Time Analysis

Visuals Used:

- **Line Chart (Transactions Count by Year) –** The line chart shows a steady increase in mobile banking transactions from 2016 (0.8K) to 2024 (7.0K), reflecting consistent growth over the years.

- **Line Chart (Year-over-Year Growth %)** – The chart shows rapid growth in mobile banking transactions between 2016 (89%) and 2018 (202%), followed by a steady decline with minor fluctuations, reaching 29% in 2024.

- **Line / Column Chart (Transaction Amount by Month-Year) –** The line chart shows a sharp upward trend in transaction amounts from 2016 (5M) to 2024 (168,269M), with steady growth until 2023 followed by an exponential spike in 2024.

- **Line Chart (Count of Transactions by Year):** Compares yearly transaction volumes and includes forecasting for the next 10 years.

- **Card Visual** – Total Transactions: Displays the overall transaction count as a key KPI.

- **Card Visual – YoY Growth %:** Shows year-over-year growth percentage for performance comparison.

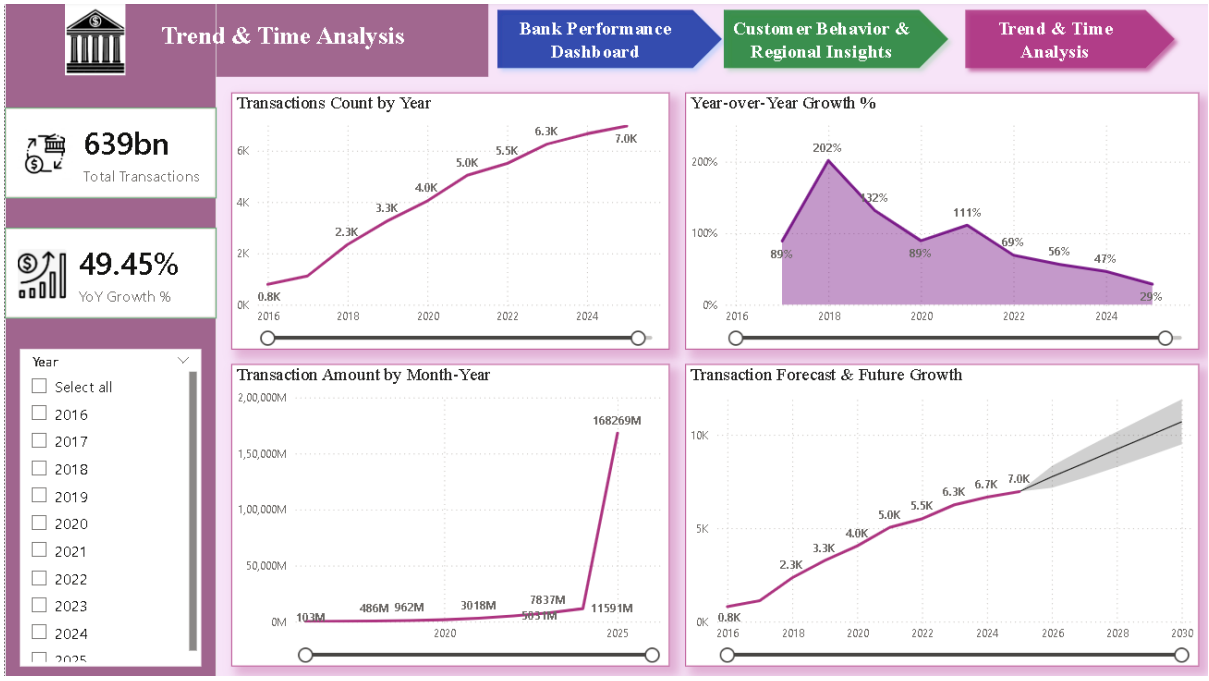
Interactive Features:

- **Year Slicer:** Allows filtering by specific years to analyze trends interactively.

- **Button Navigation** (custom buttons for page switching)

- **Edit Interactions** (controlled slicer impact on visuals and cards)

- **Consistent Layout and Color Theme**



11. Insights & Conclusions:

Bank Performance Analysis

- State Bank of India leads with ~167B transactions, followed by HDFC Bank (~55.7B) and Bank of Baroda (~41.4B), showing that a few top banks dominate transaction volume, while most other banks contribute significantly lower volumes.

- Public Sector Banks lead with ~8.4B active customers, followed by Private Sector Banks (~4.0B) and Payment Banks (~1.6B), while other bank types have significantly lower and niche customer bases.

- State Bank f India leads with ~5.66B active customers, followed by HDFC Bank (~1.13B), while Kotak Mahindra Bank (~0.76B), ICICI Bank (~0.73B), and Axis Bank (~0.72B) form the next tier, indicating a highly concentrated customer base among top banks.

- Efficiency analysis shows SBI as both a high-volume and high-value performing bank.

Customer Behavior & Regional Insights

- Maharashtra leads with (5,441) active customers, followed by Gujarat (3,681) and Karnataka (2,042), while most other states have significantly lower counts, indicating customer concentration in key economic regions.

- Large customers (~10.46B) contribute the most, far exceeding Small (~3.63B) and Medium (~0.47B) segments.

- Urban states show higher adoption rates, while rural states show steady but slower growth.
- Maharashtra (HQ ~175M) leads in average transactions, followed by Punjab (~59M) and Delhi (~44M), with a sharp drop across other states.

Trend & Time Analysis

- **Transaction count** shows steady growth from 797 in 2016 to 6,966 in 2025, indicating consistent year-over-year increase in activity, with significant acceleration after 2018.
- **YoY growth** peaked in 2018 at 201.80%, followed by strong growth in 2019 (131.95%) and 2021 (111.26%), but gradually declined to 28.78% by 2025, indicating a shift from rapid expansion to a more stable growth phase.
- Mobile banking has experienced **accelerated adoption and scaling**, with 2024 marking an unprecedented surge in transaction value — indicating both deeper customer engagement and rapid expansion of digital financial services.
- **Forecasting** indicates continued growth in mobile banking over the coming years. Both historical and forecast data confirm a consistent upward trajectory, with monthly cumulative jumps translating into long-term growth forecasts that nearly double between 2025 and 2030.

12. Conclusion:

- Mobile Banking Trends & Performance Analysis (2016–2025) highlights significant growth in digital transactions, customer adoption, and banking efficiency over time.
- The analysis provides clear insights into bank performance, regional usage patterns, and customer behavior using key KPIs.
- These findings support data-driven decision-making and indicate strong long-term growth potential in mobile banking services.